

Topic 16: Forces in air and water

Students should:		Guidance <i>iPrimary Curriculum Reference</i>
16.1	know that unsupported objects fall towards the Earth because of the force of gravity acting between the Earth and the falling object	a detailed explanation of gravity is NOT required <i>Year 6: Forces in air and water</i>
16.2	understand that weight is a force and forces are measured in newtons (N)	<i>Year 6: Forces in air and water</i>
16.3	understand that more than one force can act on an object at the same time and these forces can be represented by the direction and size of an arrow	understand forces may be balanced or unbalanced and the effect this will have on the direction and/or motion of an object <i>Year 6: Forces in air and water</i>

Students should:		Guidance
		<i>iPrimary Curriculum Reference</i>
16.4	know that the force of friction acts on moving objects to slow them down	consider examples: • where friction is useful, for example car and bicycle brakes • where friction is not useful, for example bearings. know how the effects of friction can be reduced by streamlining, smoothing surfaces and the use of oil <i>Year 6: Forces in air and water</i>
16.5	understand how friction can be used to affect how well an object grips to a surface	for example the tread of a shoe or tyre; the roughness of a road surface approaching an important crossing/junction <i>Year 6: Forces in air and water</i>
16.6	understand that friction can act between solid surfaces and air and between solid surfaces and water	<i>Year 6: Forces in air and water</i>
16.7	understand that air resistance and water resistance are forces that reduces the speed at which objects move	<i>Year 6: Forces in air and water</i>
16.8	explain how the shape of objects can be used to reduce the effects of water and air resistance, including the term <i>streamlined</i>	<i>Year 6: Forces in air and water</i>